

High-temperature operation of III-nitride high-electron-mobility transistors

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ABSTRACT

High-electron-mobility transistors (HEMTs) made with III-nitride materials are of potential use in high-temperature electronic applications including power electronics, communications, aerospace and space exploration. However, the demands of such applications make it essential to understand the thermal limits and performance evolution of III-nitride HEMTs. Here, we analyze the high-temperature operation of III-nitride HEMTs, examining the impact on material properties, device structure, and circuit-level behavior. We explore the role of critical device layers — including barrier and channel engineering, substrate selection, and passivation strategies — in mitigating high-temperature-induced effects, and evaluate the thermal stability of III-nitride HEMTs in logic, radiofrequency, and power electronics applications. We also highlight key remaining challenges in the design and optimization of III-nitride devices for high-temperature applications.

Introduction

Wide-bandgap semiconductors such as silicon carbide (SiC) and gallium nitride (GaN) are critical for high-temperature electronics. The intrinsic carrier concentration in semiconductors increases exponentially with temperature, meaning that narrow bandgap semiconductors like silicon ($E_g = 1.11$ eV) typically become unsuitable above 125°C.^{1,2} Dielectric isolation, such as in silicon-on-insulator technology, can mitigate leakage currents and bring the operating limit up to 300°C. Other conventional semiconductors like germanium ($E_g = 0.67$ eV), gallium arsenide ($E_g = 1.43$ eV), and indium phosphide ($E_g = 1.35$ eV) are similarly constrained. In contrast, GaN ($E_g = 3.4$ eV) and its III-nitride (III-N) variants offer superior thermal and electronic transport properties, both at room and high temperatures. GaN is also piezoelectric, has a direct bandgap, and can be grown heteroepitaxially on cost-effective substrates.

The most widely developed GaN device architecture is the high-electron-mobility transistor (HEMT). HEMTs exploit the formation of a two-dimensional electron gas (2DEG) at a heterojunction interface to provide high-speed, high-power operation with minimal leakage even at high temperatures. Unlike doped bulk conduction in other wide-bandgap devices, such as SiC or Ga₂O₃, the polarization-based 2DEG allows GaN HEMTs to achieve high carrier density without relying on thermally activated dopants, which can suffer from compensation and require high processing temperatures.^{3,4} GaN HEMTs were first commercialized for use in radiofrequency power amplifiers, but they now support a range of mission-critical applications where conventional silicon electronics will fail. These include high-temperature analog front-ends for radar and telemetry, logic elements for onboard control and decision-making, and integrated sensor interfaces for magnetic field detection, infrared spectroscopy, and gas monitoring in harsh environments. In power electronics, GaN HEMTs are used for efficient d.c./d.c. and d.c./a.c. conversion with low cooling overheads, enabling lighter and more compact systems in energy and transportation systems.

The applications of high-temperature III-N HEMTs can be broadly categorized into two regimes: those that involve brief exposure to ultra-high temperatures, and those that involve prolonged operation at high temperatures⁵ (**Fig. 1a**). The first regime is exemplified by hypersonic systems, where speeds can exceed Mach 5. In low Earth orbit, and even with thermal shielding, surface temperatures can reach 2,000°C and the internal electronics environments over 500°C. Therefore, hypersonic flight requires integrated sensors and high-frequency readout electronics that are capable of sustained operation above 800°C.⁶ The second regime includes terrestrial and space-based technologies. In automotives, positioning control electronics located close to the engine and gearbox can improve ignition timing, fuel efficiency, and reliability, while electric and hybrid vehicles require inverters, converters, and sensors that can operate at 200°C and last the vehicle's lifespan. Reducing the footprint of a vehicle's thermal management, which consumes up to one-third of their volume, also lowers carbon emissions.⁷ Aerospace systems similarly benefit from placing electronics closer to hot zones to reduce latency and mass while ensuring reliability